BEST KNOWN SQUARE PACKINGS

- **O** proved optimal (77)
- **=** exact value known (287)
- **≈** only known numerically (37)
- **R** rigid (22)

★ recent result, since August, 2026 (297)

90° 45° colors indicate distinct tilt angles

4 3 2 1 0 shade indicates number of full-side contacts

The square packing problem asks for the side $s(n)$ of the smallest square that can hold n unit squares, where the squares are free to rotate but cannot overlap. deg is the algebraic degree of that side length

Best packings due to Göbel, Trump, Bidwell, Hämäläinen, Wainwright, Friedman, Ellsworth, Schadt, Cantrell, DeVincentis, Stenlund, Morandi, hmbelvedere, Hajba, Chang, Couzo, Daniel, de Winter, Chaoweeraprasit.

The Squares Project

github.com/jlevy/squares

Diagram by Joshua Levy with the help of a few billion tokens from Claude and Codex

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